Tabela surowych próbek dyskretnych – Optymalizacja dwutorowa 2D PSO Trend MF-BB

Zestawienie dyskretnych próbek trajektorii czasowych (t, v_C, i_L, s) dla regulatora Trend MF-BB z nastawami z optymalizacji 2D PSO (w_i* = 0.4383, fc_i* = 428.9 Hz, w_u* = 15.61, fc_u* = 421.1 Hz, T_s = 10 us).

Okno 1: Stan ustalony pod pełnym obciążeniem (240 V, 1.2 kW, t □ [145.0, 146.5] ms, krok 20 µs)

t [ms]	v_C [V]	i_L [A]	s	|	t [ms]	v_C [V]	i_L [A]	s
145.000	239.73	12.977	1	|	145.760	239.48	14.516	1
145.020	239.89	12.430	0	|	145.780	240.24	10.769	0
145.040	239.97	11.880	1	|	145.800	240.23	10.218	1
145.060	239.46	14.529	1	|	145.820	239.73	12.869	1
145.080	240.23	10.783	0	|	145.840	239.89	12.322	0
145.100	240.22	10.232	1	|	145.860	239.96	11.773	1
145.120	239.72	12.883	1	|	145.880	239.46	14.422	1
145.140	239.88	12.336	0	|	145.900	240.21	10.676	0
145.160	239.95	11.787	1	|	145.920	239.71	13.327	1
145.180	239.44	14.437	1	|	145.940	239.89	12.779	0
145.200	240.20	10.691	0	|	145.960	240.48	9.026	0
145.220	239.70	13.341	1	|	145.980	239.98	11.679	1
145.240	239.88	12.794	0	|	146.000	239.48	14.328	1
145.260	240.47	9.041	0	|	146.020	240.22	10.582	0
145.280	239.97	11.694	1	|	146.040	239.72	13.232	1
145.300	239.47	14.343	1	|	146.060	239.90	12.685	0
145.320	240.21	10.597	0	|	146.080	240.48	8.931	0
145.340	239.71	13.247	1	|	146.100	239.97	11.585	1
145.360	239.89	12.700	0	|	146.120	239.47	14.234	1
145.380	240.47	8.947	0	|	146.140	240.21	10.488	0
145.400	239.97	11.600	1	|	146.160	239.70	13.139	1
145.420	239.47	14.249	1	|	146.180	239.88	12.592	0
145.440	240.20	10.503	0	|	146.200	239.96	12.042	1
145.460	239.70	13.154	1	|	146.220	239.46	14.691	1
145.480	239.88	12.607	0	|	146.240	240.24	10.944	0
145.500	239.96	12.058	1	|	146.260	240.24	10.393	1
145.520	239.46	14.707	1	|	146.280	239.73	13.044	1
145.540	240.24	10.960	0	|	146.300	239.90	12.497	0
145.560	240.24	10.408	1	|	146.320	239.98	11.947	1
145.580	239.73	13.059	1	|	146.340	239.48	14.596	1
145.600	239.91	12.512	0	|	146.360	240.25	10.849	0
145.620	239.98	11.962	1	|	146.380	240.24	10.297	1
145.640	239.48	14.611	1	|	146.400	239.74	12.949	1
145.660	240.25	10.864	0	|	146.420	239.91	12.401	0
145.680	240.25	10.313	1	|	146.440	239.98	11.852	1
145.700	239.74	12.964	1	|	146.460	239.47	14.501	1
145.720	239.91	12.416	0	|	146.480	240.24	10.754	0
145.740	239.98	11.867	1	|	146.500	240.23	10.203	1

Okno 2: Stan przejściowy – skok zadanego napięcia 240 V → 160 V (t □ [149.8, 151.3] ms, krok 20 µs)

t [ms]	v_C [V]	i_L [A]	s	|	t [ms]	v_C [V]	i_L [A]	s
149.800	239.88	12.449	0	|	150.560	206.13	-18.326	1
149.820	239.96	11.900	1	|	150.580	203.73	-21.099	0
149.840	239.45	14.549	1	|	150.600	202.22	-21.112	1
149.860	240.22	10.802	0	|	150.620	201.79	-18.419	1
149.880	240.21	10.251	1	|	150.640	199.40	-21.076	0
149.900	239.71	12.902	1	|	150.660	197.90	-21.031	1
149.920	239.87	12.356	0	|	150.680	197.48	-18.338	1
149.940	239.94	11.807	1	|	150.700	195.11	-20.881	0
149.960	239.44	14.456	1	|	150.720	193.63	-20.780	1
149.980	240.20	10.710	0	|	150.740	193.23	-18.087	1
150.000	239.69	13.361	1	|	150.760	190.90	-20.518	0
150.020	240.34	9.611	0	|	150.780	189.44	-20.360	1
150.040	240.62	5.854	0	|	150.800	189.04	-17.668	1
150.060	240.51	2.099	0	|	150.820	186.77	-19.989	0
150.080	240.04	-1.643	0	|	150.840	184.27	-22.242	0
150.100	239.19	-5.362	0	|	150.860	183.88	-19.548	1
150.120	237.97	-9.049	0	|	150.880	182.56	-19.298	0
150.140	236.39	-12.694	0	|	150.900	180.15	-21.442	0
150.160	234.45	-16.288	0	|	150.920	179.77	-18.748	1
150.180	232.16	-19.820	0	|	150.940	178.50	-18.445	0
150.200	229.52	-23.282	0	|	150.960	176.18	-20.483	0
150.220	229.04	-20.586	1	|	150.980	174.76	-20.130	1
150.240	228.56	-17.894	1	|	151.000	174.40	-17.438	1
150.260	226.13	-21.265	0	|	151.020	172.20	-19.369	0
150.280	224.55	-21.575	1	|	151.040	169.81	-21.237	0
150.300	224.08	-18.882	1	|	151.060	169.45	-18.543	1
150.320	222.70	-19.168	0	|	151.080	169.10	-15.854	1
150.340	220.16	-22.380	0	|	151.100	168.74	-13.168	1
150.360	219.69	-19.685	1	|	151.120	166.98	-14.960	0
150.380	218.28	-19.912	0	|	151.140	166.63	-12.275	1
150.400	215.68	-23.004	0	|	151.160	166.28	-9.593	1
150.420	215.23	-20.309	1	|	151.180	165.50	-9.126	0
150.440	214.78	-17.617	1	|	151.200	164.68	-8.649	1
150.460	212.42	-20.622	0	|	151.220	164.33	-5.973	1
150.480	210.91	-20.751	1	|	151.240	163.99	-3.300	1
150.500	210.47	-18.059	1	|	151.260	163.23	-4.991	0
150.520	208.08	-20.948	0	|	151.280	162.62	-4.491	1
150.540	206.56	-21.019	1	|	151.300	162.28	-1.820	1

Okno 3: Stan ustalony po skoku pod obciążeniem (160 V, 538 W, t □ [170.0, 171.5] ms, krok 20 µs)

t [ms]	v_C [V]	i_L [A]	s	|	t [ms]	v_C [V]	i_L [A]	s
170.000	159.95	5.985	1	|	170.760	160.13	4.546	1
170.020	159.96	6.511	0	|	170.780	159.79	7.204	1
170.040	160.19	4.901	0	|	170.800	160.10	5.597	0
170.060	160.08	5.426	1	|	170.820	160.24	3.986	0
170.080	160.06	5.951	0	|	170.840	159.90	6.646	1
170.100	160.24	4.339	0	|	170.860	160.15	5.037	0
170.120	160.11	4.864	1	|	170.880	160.24	3.425	0
170.140	159.77	7.522	1	|	170.900	159.91	6.086	1
170.160	160.11	5.915	0	|	170.920	159.92	6.612	0
170.180	160.28	4.302	0	|	170.940	160.17	5.003	0
170.200	160.14	4.826	1	|	170.960	160.26	3.391	0
170.220	159.81	7.485	1	|	170.980	159.92	6.052	1
170.240	160.14	5.876	0	|	171.000	159.93	6.578	0
170.260	160.32	4.263	0	|	171.020	160.18	4.969	0
170.280	160.17	4.787	1	|	171.040	160.26	3.357	0
170.300	159.84	7.445	1	|	171.060	159.92	6.017	1
170.320	160.17	5.836	0	|	171.080	159.93	6.544	0
170.340	160.33	4.222	0	|	171.100	160.17	4.934	0
170.360	160.19	4.746	1	|	171.120	160.07	5.459	1
170.380	159.85	7.404	1	|	171.140	160.05	5.984	0
170.400	160.18	5.795	0	|	171.160	160.23	4.373	0
170.420	160.34	4.181	0	|	171.180	160.10	4.897	1
170.440	160.20	4.704	1	|	171.200	159.76	7.556	1
170.460	159.86	7.363	1	|	171.220	160.10	5.948	0
170.480	160.18	5.753	0	|	171.240	160.28	4.336	0
170.500	160.34	4.139	0	|	171.260	160.14	4.860	1
170.520	160.19	4.663	1	|	171.280	159.81	7.519	1
170.540	159.86	7.322	1	|	171.300	160.14	5.910	0
170.560	160.18	5.712	0	|	171.320	160.32	4.297	0
170.580	160.33	4.098	0	|	171.340	160.18	4.820	1
170.600	160.18	4.622	1	|	171.360	160.13	5.345	0
170.620	159.85	7.281	1	|	171.380	160.25	3.733	0
170.640	160.16	5.672	0	|	171.400	159.92	6.393	1
170.660	160.31	4.059	0	|	171.420	159.95	6.919	0
170.680	160.16	4.583	1	|	171.440	160.22	5.309	0
170.700	159.82	7.242	1	|	171.460	160.34	3.695	0
170.720	160.13	5.633	0	|	171.480	160.00	6.355	1
170.740	160.28	4.021	0	|	171.500	160.22	4.744	0